

XRPL Validator — Daily Engineering Report · 2026-08-11

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- 7 commits (xrpl-validator 7)

Engineering changes

- **fix(xrpl-ledger): real LP token math for two-asset deposits and all withdrawals** (c9ac386 , xrpl-validator)
Removes the last of the 1e7 LPToken placeholders from the deposit/withdraw paths. AMMWithdraw 7 -> 0 hits, AMMDeposit 7 -> 3; the AMM value family is now 3, from 23 at gate 46.
- **fix(xrpl-ledger): a single-asset deposit pays what its adjusted tokens are worth** (5197c37 , xrpl-validator)
CLOSES THE AMM VALUE FAMILY: 23 hits at gate 46 -> 0. #105813899 2C8C37F9 was the last one, 55/55 with no DX_VALCHECK output at all.
- **fix(xrpl-ledger): a fully-consumed offer transfers its own amounts verbatim** (dd4991a , xrpl-validator)
OfferCreate value hits 48 -> 26 across the ledgers carrying them, with the XRP family almost gone (AccountRoot.Balance 18 -> 2). No mutation-set regressions.
- **fix(xrpl-ledger): a payment charges the input issuer's rate on EVERY hop** (5c16b88 , xrpl-validator)
Payment value hits 12 -> 4. #105924683 B3AA3ACC — the largest remaining error in the whole backlog at 2.96e-3, 5 hits — closes outright: 73/73 with no DX_VALCHECK output.
- **fix(xrpl-ledger): a bridged taker pays the input issuer's transfer fee** (94a028e , xrpl-validator) Closes the two largest remaining OfferCreate errors. #105887283 4A13D048 goes 88/88 with no DX_VALCHECK output at all; #105954798 D5887FD7 drops from 1.34e-4 to 2.0e-11.
- **fix(xrpl-ledger): to_amount honours round-to-NEAREST instead of flooring it** (6de2684 , xrpl-validator)
Closes the largest remaining cluster: #105843839 C1F9FB1F, 6 hits -> 0. That ledger now carries one pre-existing 1.5e-11 Payment residue and nothing else, 125/125.
- **fix(xrpl-ledger): an XRP residual ceils to whole DROPS, not to 16 digits** (df3c8f2 , xrpl-validator)
#105672435 B409D45C -> 0 hits, 118/118. It was the largest genuine error left (6.05e-6) and the last one above 2e-6.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Oracle-Verified Account Spend Caps** (accounts) Add an AccountPolicy ledger object (owner-reserved, keylet::accountPolicy(account)) set by a new AccountPolicySet transaction, carrying sfMaxOutboundV
- **Credential-Gated Inbound Firewall Rules** (compliance) Add an IsfFirewall flag settable via AccountSet plus a bounded FirewallRule directory (ltFIREWALL_RULE, owner-reserved, keylet::firewallRule(account,
- **Sponsored Multi-Destination Remittance Payments** (payments) Implement ttREMIT per XLS-55: a single transaction carrying a bounded STArray sfAmounts (max 8 entries) of mixed XRP/IOU/MPT amount-destination pairs,

Appendix — commits

Time	Repo	Commit	Subject
09:20	xrpl-validator	c9ac386	fix(xrpl-ledger): real LP token math for two-asset deposits and all withdrawals
10:16	xrpl-validator	5197c37	fix(xrpl-ledger): a single-asset deposit pays what its adjusted tokens are worth
11:50	xrpl-validator	dd4991a	fix(xrpl-ledger): a fully-consumed offer transfers its own amounts verbatim
13:53	xrpl-validator	5c16b88	fix(xrpl-ledger): a payment charges the input issuer's rate on EVERY hop
16:07	xrpl-validator	94a028e	fix(xrpl-ledger): a bridged taker pays the input issuer's transfer fee
18:36	xrpl-validator	6de2684	fix(xrpl-ledger): to_amount honours round-to-NEAREST instead of flooring it
22:18	xrpl-validator	df3c8f2	fix(xrpl-ledger): an XRP residual ceils to whole DROPS, not to 16 digits